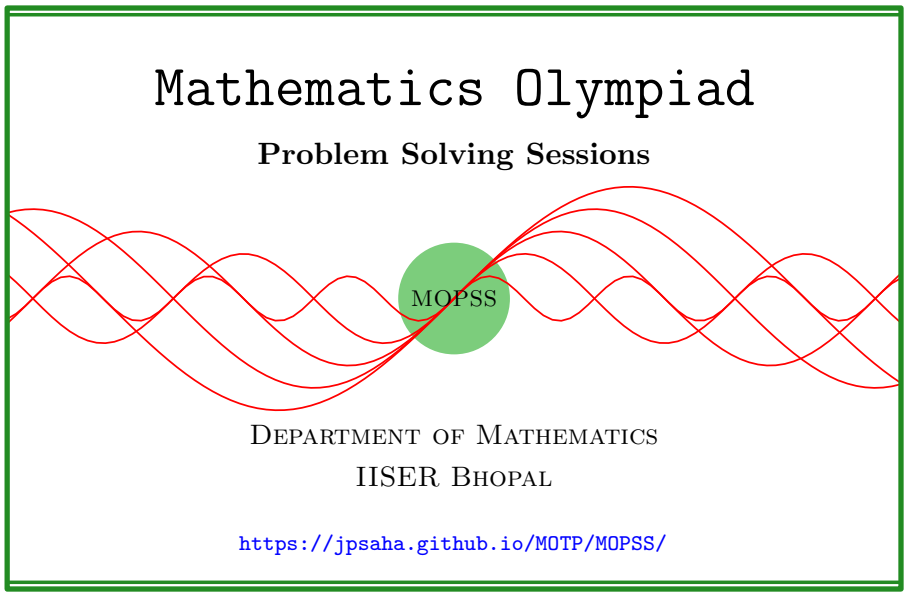
Warm up

MOPSS



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

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§1 Warm up

Exercise 1 (All-Russian Mathematical Olympiad 2014 Grade 9 Day 1 P1, , by S. Berlov)**.** On a circle, there are 99 positive integers. If a, b are any two neighbouring numbers on the circle, then $a - b$ is equal to 1 or 2 or $\frac{a}{b} = 2$. Prove that there exists a natural number on the circle that is divisible by 3.

> **Walkthrough** —
>
> **(a)** Show that if none of the numbers on the circle is divisible by 3, then all the numbers on the circle are congruent to either 1 or 2 modulo 3.
>
> **(b)** Using that 99 is odd, show that it is impossible to arrange 99 numbers on the circle such that none of them is divisible by 3 and any two neighbouring numbers are not congruent modulo 3.

Solution 1. On the contrary, let us assume that none of the numbers on the circle is divisible by 3. Then, each number on the circle is congruent to either 1 or 2 modulo 3. The given conditions imply that any two neighbouring numbers on the circle are not congruent modulo 3. Since there are 99 numbers on the circle and 99 is odd, it is impossible to have such a configuration. ■

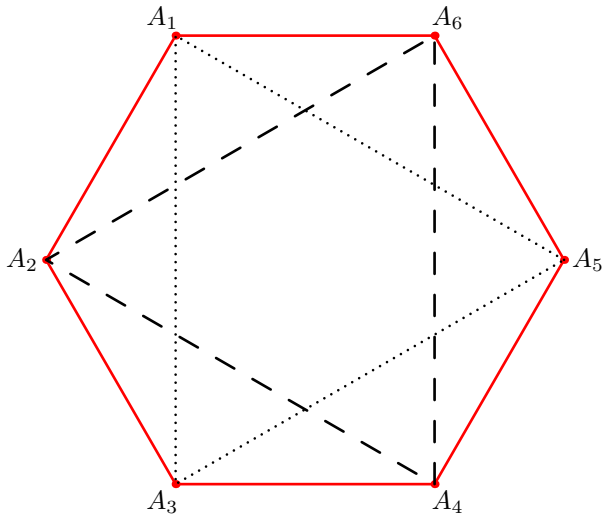


Figure 1: India IOQM 2023 P16, Exercise 2

Remark. If none of the numbers on the circle is divisible by 3, then using their congruence classes modulo 3 yields a 2-coloring of a 99 cycle such that no two adjacent vertices are of the same color, implying that[a] the cycle is bipartite. This is a contradiction since odd cycles are not bipartite.

[a]It also implies that the odd 99 cycle has chromatic number 2, which is impossible.

Exercise 2 (IOQM 2023 P16, , cf. ARML 2010 Team Problems P4)**.** The six sides of a convex hexagon $A_1A_2A_3A_4A_5A_6$ are colored red. Each of the diagonals of the hexagon is colored either red or blue. If N is the number of such colorings such that every triangle $A_iA_jA_k$, where $1 \leq i < j < k \leq 6$, has at least one red side, find the sum of the squares of the digits of N.

Summary — It suffices to (observe and then!!) show that for a coloring of the diagonals of the hexagon using red and blue, if each of the triangles $A_1A_2A_3$ and $A_2A_4A_6$ has at least one red side, then each of the triangles $A_iA_jA_k$, where $1 \leq i < j < k \leq 6$, has at least one red side too. Count the number of colorings of the diagonals of the hexagon using red and blue such that each of the triangles $A_1A_3A_5$ and $A_2A_4A_6$ has at least one red side.

Walkthrough —

(a) The sides of the hexagon has been colored red.

(b) Note that if $A_iA_jA_k$ is such that some two of its vertices are consecutive, then it has at least one red side.

(c) If $A_iA_jA_k$ is such that no two of its vertices are consecutive, then any two of its vertices are exactly one vertex apart. Note that there precisely two such triangles, namely, $A_1A_3A_5$ and $A_2A_4A_6$.

(d) Observe that if each of these two triangles have at least one red side, then the required condition is satisfied.

(e) Moreover, if the required condition is satisfied, then each of these two triangles has at least one red side.

(f) It follows that N is equal to the number of colorings of the diagonals of the hexagon using red and blue such that each of the triangles $A_1A_2A_3$ and $A_2A_4A_6$ have at least one red side.

(g) This shows that

$$N = (2^3 - 1)(2^3 - 1)(2^{\binom{6}{2}-6-(3+3)}) = 7 \cdot 7 \cdot 8 = 392.$$

(h) The sum of the squares of the digits of N is equal to $3^2 + 9^2 + 2^2 = 9 + 81 + 4 = 94$.

Solution 2. Consider one coloring of the diagonals of the hexagon using red and blue. If this coloring satisfies the given condition, then each of the triangles $A_1A_2A_3$ and $A_2A_4A_6$ has at least one red side. Let us establish the claim below.

Claim — For any coloring of the diagonals of the hexagon using red and blue, if each of the triangles $A_1A_2A_3$ and $A_2A_4A_6$ has at least one red side, then each of the triangles $A_iA_jA_k$, where $1 \leq i < j < k \leq 6$, has at least one red side.

Proof of the Claim. Consider a triangle $A_iA_jA_k$ with $1 \leq i < j < k \leq 6$. Note that if $A_iA_jA_k$ is such that some two of its vertices are consecutive, then it has at least one red side. If $A_iA_jA_k$ is such that no two of its vertices are consecutive, then any two of its vertices are exactly one vertex apart. Note that there precisely two such triangles, namely, $A_1A_3A_5$ and $A_2A_4A_6$. This proves the Claim. □

Hence, it follows that N is equal to the number of colorings of the diagonals of the hexagon using red and blue such that each of the triangles $A_1A_2A_3$ and $A_2A_4A_6$ have at least one red side. Note that the sides of $A_1A_2A_3$ can be colored in $2^3 - 1 = 7$ ways such that at least one of its sides is red. Similarly, the sides of $A_2A_4A_6$ can be colored in $2^3 - 1 = 7$ ways such that at least one of its sides is red. Note that there are precisely $\binom{6}{2} - 6 - 3 - 3$ diagonals in the hexagon which are not a side of these two triangles. Hence, the number of ways to color these 15 diagonals using red and blue satisfying the given condition is equal to

$$N = (2^3 - 1)(2^3 - 1)(2^{\binom{6}{2}-6-(3+3)}) = 7 \cdot 7 \cdot 2^{15} = 392.$$

The sum of the squares of the digits of N is equal to

$$3^2 + 9^2 + 2^2 = 9 + 81 + 4 = 94.$$

■

Exercise 3 (SMMC 2020 A1,). There are 1001 points in the plane such that no three are collinear. The points are joined by 1001 line segments such that each point is an endpoint of exactly two of the line segments. Prove that there does not exist a straight line in the plane that intersects each of the 1001 line segments in an interior point. An **interior point** of a line segment is a point of the line segment that is not one of the two endpoints.

Walkthrough —

(a) On the contrary, let us assume that there exists a straight line L in the plane that intersects each of the 1001 line segments in an interior point.

(b) Show that L cannot pass through any of the 1001 points.

(c) Color the points on one side of L red, and the points on the other side of L blue.

(d) Show that the number of the line segments is equal to twice the number of red points, to obtain a contradiction.

Solution 3. On the contrary, let us assume that there exists a straight line L in the plane that intersects each of the 1001 line segments in an interior point. Note that L cannot pass through any of the 1001 points. Indeed, if L passes through one of the 1001 points, then the two line segments that have this point as an endpoint, cannot intersect L at their interior points, since no three of the given points are collinear. Let us color the points on one side of L red, and the points on the other side of L blue. Since L intersects each of the 1001 line segments in an interior point, each of the 1001 line segments has one red endpoint and one blue endpoint. Moreover, each of the 1001 points is an endpoint of exactly two of the line segments. Therefore, the number of line segments is equal to twice the number of red points, which is impossssible since 1001 is odd. This completes the proof. ■

Exercise 4 (RMO 2025a P1,).

(a) Let $n \geq 3$ be an integer. Find a configuration of n lines in the plane which has exactly

(i) $n - 1$ distinct points of intersection;

(ii) n distinct points of intersection.

(b) Give configurations of n lines that have exactly $n + 1$ distinct points of intersection for (i) $n = 8$ and (ii) $n = 9$.

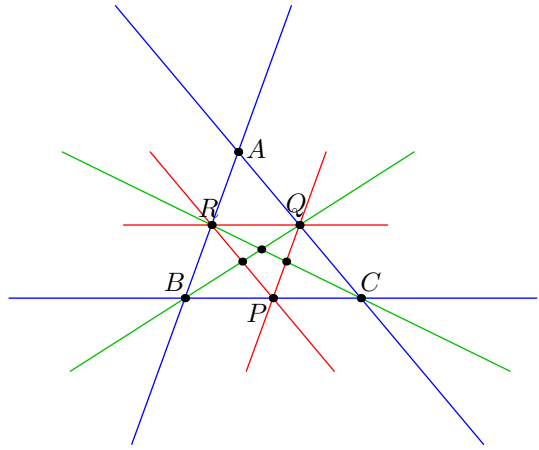


Figure 2: RMO 2025a P1, Exercise 4(b)(i): Configuration with 8 lines and 9 intersection points.

Walkthrough —

(a) For part (a)(i), consider $n-1$ parallel lines and another line perpendicular to all of them.

(b) For part (a)(ii), consider $n-1$ lines concurrent at a point and another line not passing through that point.

(c) For part (b)(ii), consider three lines forming a triangle, together with the three lines forming its medians, and three lines forming the medial triangle.

(d) For part (b)(i), consider the above configuration and remove one median.

Solution 4. Consider the configuration of $n-1$ parallel lines and another line $\mathscr{L}$ which is perpendicular to some of these parallel lines. Note that $\mathscr{L}$ is also perpendicular to all the other parallel lines. It follows that this is a configuration as required in (a)(i) since all the parallel lines intersect the line perpendicular to them at distinct points, giving exactly $n-1$ points of intersection.

For (a)(ii), consider $n-1$ lines concurrent at a point O. Next, consider a point P other than O. We will specify a line passing through P and show that this line together with the $n-1$ lines concurrent at O gives a configuration as required in (a)(ii). Note that there is only one line passing through O and P. Also note that there are precisely $n-1$ lines passing through P, such that each of these lines is parallel to one of the $n-1$ lines concurrent at O. Let $\mathscr{L}$ denote a line passing through P which is not parallel to any of the $n-1$ lines concurrent at O, and which does not pass through O. Note that $\mathscr{L}$ intersects each of the $n-1$ lines concurrent at O at a point distinct from

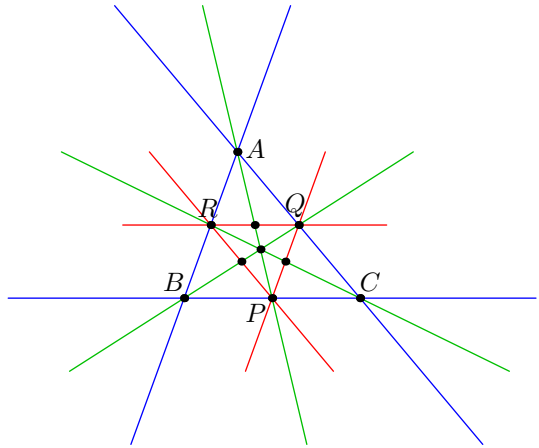


Figure 3: RMO 2025a P1, Exercise 4(b)(ii): Configuration with 9 lines and 10 intersection points.

O, and $\mathscr{L}$ intersects the $n-1$ distinct lines concurrent at O at $n-1$ distinct points. Hence, the configuration of $n-1$ lines concurrent at O together with the line $\mathscr{L}$ gives a configuration as required in (a)(ii) with exactly n points of intersection.

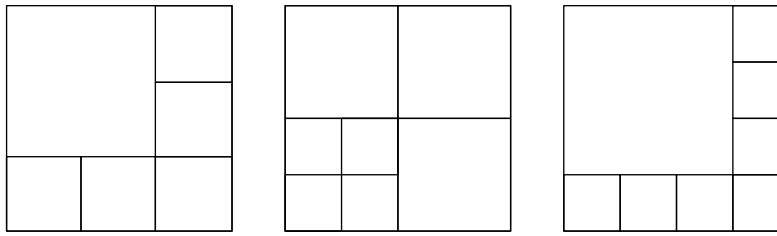
For (b)(i), consider the configuration shown in Fig. 2. Here, we have 8 lines and exactly 9 points of intersection.

For (b)(ii), consider the configuration shown in Fig. 3. Here, we have 9 lines and exactly 10 points of intersection.

More precisely, we begin with 3 lines such that no two are parallel and no three are concurrent, and their 3 points of intersection are the vertices of a triangle ABC as in Fig. 3. Let P denote the midpoint of side BC, Q the midpoint of side CA, and R the midpoint of side AB. Next, we draw the line passing through P, Q, the line passing through Q, R, and the line passing through R, P. This gives a total of 6 lines with exactly 6 points of intersection. Indeed, the 3 original lines intersect at 3 points, and PQ, QR, RP are parallel to AB, BC, CA respectively, so there are precisely 6 points of intersection, which are the vertices of triangles ABC and PQR.

Next, we draw the line passing through A, P, the line passing through B, Q, and the line passing through C, R. This gives a total of 9 lines. Note that PQ is parallel to AB, and R is the midpoint of AB, showing that $PQAR$ and $PQRB$ are parallelograms. Hence, AP, RQ bisect each other, and BQ, RP also bisect each other. Similarly, CR, PQ bisect each other. Observing that AP, BQ, CR are the medians of $\triangle ABC$, it follows that they concur at the centroid of $\triangle ABC$. This gives a total of 9 lines with exactly 10 points of intersection, as required for (b)(ii).

To obtain a configuration for (b)(i), we remove one of the lines AP, BQ, CR



(a) Tiling using 6 squares. (b) Tiling using 7 squares. (c) Tiling using 8 squares.

Figure 4: Tiling a square using 6, 7, and 8 squares.

from the above configuration. ■

Exercise 5 (RMO 2016g P1,). Suppose in a given collection of 2016 integers, the sum of any 1008 integers is positive. Show that sum of all 2016 integers is positive.

> **Walkthrough** — Think about it!

Example 6. For any integer $n \geq 6$, show that a square can be cut into n squares, not necessarily of the same size.

> **Walkthrough** —
>
> **(a)** Try to work it out for small values of n, for instance $n = 6, 7, 8, 9$.
>
> **(b)** What happens if you want to go from n to $n + 3$?

Solution 5. Note that if we can do it for n, then we can do it for $n + 3$ by cutting one of the squares into four smaller squares. So it suffices to do it for $n = 6, 7, 8$, which is shown in Fig. 4. By induction, we are done. ■

Example 7. Can a 12×15 chessboard be covered by non-overlapping trominos? A tromino is an L-shaped piece consisting of three squares, where these squares are of the same size as the smallest squares in the chessboard.

> **Walkthrough** —
>
> **(a)** Does Fig. 5 help?

Solution 6. Note that a 2×3 rectangle can be tiled by two non-overlapping trominos. Since a 12×15 chessboard can be covered by non-overlapping 2×3 rectangles, it follows that a 15×15 chessboard can be tiled by non-overlapping trominos. ■

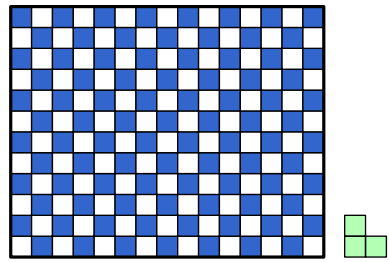


Figure 5: Tiling a 12×15 chessboard using trominos.

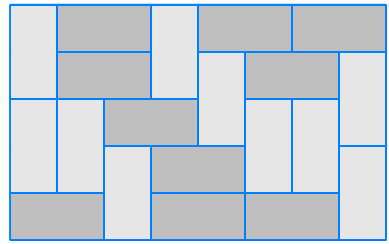


Figure 6: India BMath 2006 (a tiling of a 5×8 rectangle with non-overlapping dominoes), Example 9

Exercise 8 (SMMC 2018 A1)**.** Call a rectangle *dominant* if it is similar to a 2×1 rectangle. For which integers $n \geq 5$ is it possible to tile a square with n dominant rectangles, which are not necessarily congruent to each other?

Walkthrough —

(a)

Solution 7. ■

Example 9 (India BMath 2006)**.** A domino is a 2 by 1 rectangle. For what integers m and n, can one cover an m by n rectangle with non-overlapping dominoes?

Walkthrough —

(a) If an $m \times n$ rectangle admits a covering by non-overlapping dominos, then show that at least one of the integers m, n has to be even.

(b) If at least one of m, n is even, then prove that an $m \times n$ rectangle admits a covering by non-overlapping dominos.

Solution 8. In the following, an $m \times n$ rectangle is to be thought as an $m \times n$ rectangular grid.

To be able to cover an $m \times n$ rectangle by non-overlapping dominoes, it is necessary for the product mn to be even, and hence, at least one of m, n is even. Indeed, if an $m \times n$ rectangle admits a covering using k non-overlapping dominoes, then those dominoes together cover $2k$ unit squares, and this yields that $2k = mn$.

Moreover, when at least one of m, n is even, an $m \times n$ rectangle can be covered by non-overlapping dominoes by covering each row by $m/2$ (resp. each column by $n/2$) non-overlapping dominos if m (resp. n) is even.

This shows that an $m \times n$ rectangle can be covered by non-overlapping dominoes if and only if at least one of m, n is even. ■

> **Remark.** The above conclusion shows that an $m \times n$ rectangle admits a covering by non-overlapping dominoes if and only if it admits a covering by non-overlapping dominoes in the *most obvious manner*, i.e. a covering by non-overlapping dominoes such that all of them are either horizontal or vertical (cf. [Bru10, p. 6]).

The following problem is a more general version of Example 9.

Exercise 10. [Eng98, Problem 8, Chapter 2, p. 26] Show that an $m \times n$ rectangle admits a covering by non-overlapping $k \times 1$ rectangles if and only if k divides m or k divides n.

Exercise 11. [Eng98] [Bru10, p. 4] Consider an $n \times n$ chessboard and remove the squares at the end of one diagonal. Determine whether the mutilated chessboard admits a covering by non-overlapping dominos.

Exercise 12 (Tournament of Towns Fall 2013, Senior A Level P4,). Integers $1, 2, \ldots, 100$ are written on a circle, not necessarily in that order. Can it be that the absolute value of the difference between any two adjacent integers is at least 30 and at most 50?

> **Walkthrough** —
>
> **(a)** Assume that such an arrangement is possible.
>
> **(b)** Consider the integers $1, 2, \ldots, 25, 76, 77, \ldots, 100$.
>
> **(c)** Show that no two of these integers are adjacent.
>
> **(d)** Conclude that the integers $1, 2, \ldots, 25, 76, 77, \ldots, 100$ and $26, 27, \ldots, 75$ are placed alternately along the circle.
>
> **(e)** Show that 26 is adjacent to 76 only.
>
> **(f)** Derive a contradiction.

Solution 9. Let us assume that the integers $1, 2, \ldots, 100$ can be arranged along the circumference of a circle in some order such that the absolute value of the difference between any two adjacent integers is at least 30 and at most 50.

Since the difference of any two of the integers $1, 2, \ldots, 25, 76, 77, \ldots, 100$ is less than 30 or greater than 50 it follows that no two of these integers are adjacent. Consequently, the elements of the sets

$$\{1, 2, \ldots, 25, 76, 77, \ldots, 100\}, \{26, 27, \ldots, 75\}$$

are placed alternately along the circle. However, 26 is adjacent to 26 only, which is impossible. This shows that there is no arrangement of the integers $1, 2, \ldots, 100$ along the circumference of a circle satisfying the given condition. ■

Exercise 13 (RMO 2003 P7,). Consider the set $X = \{1, 2, 3, \ldots, 9, 10\}$. Find two disjoint nonempty subsets A and B of X such that

(a) $A \cup B = X$,

(b) $\text{prod}(A)$ is divisible by $\text{prod}(B)$, where for any finite set of numbers C, $\text{prod}(C)$ denotes the product of all numbers in C,

(c) the quotient $\text{prod}(A)/\text{prod}(B)$ is as small as possible.

Summary — It is equivalent to finding a subset B of $\{1, \ldots, 10\}$, other than $\emptyset, \{1, \ldots, 10\}$, such that $\text{prod}(B)^2$ divides $10!$ and the quotient $10!/\text{prod}(B)^2$ is minimized. To do so,

(a) write down the prime power factorization of $10!$,

(b) throw in enough elements in B so that $\text{prod}(B)$ is maximized, and $\text{prod}(B)^2$ divides $10!$.

Walkthrough —

(a) Observe that it is enough to find a nonempty proper subset B of $\{1, 2, \ldots, 10\}$ such that $\text{prod}(B)^2$ divides $10!$ and $\text{prod}(B)$ is the maximum.

(b) Writing down the prime power factorization of $10!$, deduce that B does not contain 7, it contains a multiple of 5, and also a multiple of 2 and a multiple of 3.

(c) Prove that B contains exactly one multiple of 5, and not more that two multiples of 3.

(d) Show that B is equal to one of the subsets $\{5, 3, 6, 2^3\}$, $\{5, 3, 6, 2^3, 1\}$, $\{5, 3, 6, 2, 2^2\}$, $\{5, 3, 6, 2, 2^2, 1\}$, $\{5, 9, 2, 2^3\}$, $\{5, 9, 2, 2^3, 1\}$, $\{10, 3, 6, 2^2\}$, $\{10, 3, 6, 2^2, 1\}$, $\{10, 9, 2^3\}$, $\{10, 9, 2^3, 1\}$, $\{10, 9, 2^3\}$, $\{10, 9, 2^3, 1\}$.

(e) Show that any of these three subsets also have the stated property.

First, let's work on it. Let A, B be two nonempty disjoint subsets of X satisfying the required conditions (note that such subsets exist since X can be written as the union of two disjoint subsets in finitely many ways only). Due to the equality

$$\frac{\text{prod}(A)}{\text{prod}(B)} = \frac{10!}{(\text{prod}(B))^2},$$

it is equivalent to having a subset B of X such that $\text{prod}(B)^2$ divides $10!$ and $\text{prod}(B)$ is the maximum. Note that $10!$ is equal to the product $2^8 \cdot 3^4 \cdot 5^2 \cdot 7$. So $\text{prod}(B)$ divides $2^4 \cdot 3^2 \cdot 5$, and hence, B does not contain 7. Moreover, B contains a multiple of 5, otherwise $(\text{prod}(B \cup \{5\}))^2$ would divide $10!$ and $\text{prod}(B \cup \{5\})$ would be strictly larger than $\text{prod}(B)$, which contradicts the choice of B. Similarly, B also contains a multiple of 2 and a multiple of 3. Note that B contains exactly one multiple of 5 (since $5^3 \nmid 10!$). Since $(\text{prod}(B))^2$ divides $10!$ and $\text{prod}(B)$ is the maximum, B is equal to one of the following sets

- $\{5, 3, 2, 2^3\}$, $\{5, 3, 2, 2^3, 1\}$, $\{5, 6, 2, 2^3\}$, $\{5, 6, 2, 2^3, 1\}$, $\{5, 3, 6, 2^3\}$, $\{5, 3, 6, 2^3, 1\}$, $\{5, 3, 6, 2, 2^2\}$, $\{5, 3, 6, 2, 2^2, 1\}$, $\{5, 9, 2, 2^3\}$, $\{5, 9, 2, 2^3, 1\}$ if B contains 5,
- $\{10, 3, 2^3\}$, $\{10, 3, 2^3, 1\}$, $\{10, 3, 2, 2^2\}$, $\{10, 3, 2, 2^2, 1\}$, $\{10, 6, 2^2\}$, $\{10, 6, 2^2, 1\}$, $\{10, 3, 6, 2^2\}$, $\{10, 3, 6, 2^2, 1\}$, $\{10, 9, 2^3\}$, $\{10, 9, 2^3, 1\}$, $\{10, 9, 2, 2^2\}$, $\{10, 9, 2, 2^2, 1\}$ if B contains 10.

For any of the above sets, the product of its elements is equal to 240, 480, or 720. So B is equal to one of the sets $\{5, 3, 6, 2^3\}$, $\{5, 3, 6, 2^3, 1\}$, $\{5, 3, 6, 2, 2^2\}$, $\{5, 3, 6, 2, 2^2, 1\}$, $\{5, 9, 2, 2^3\}$, $\{5, 9, 2, 2^3, 1\}$, $\{10, 3, 6, 2^2\}$, $\{10, 3, 6, 2^2, 1\}$, $\{10, 9, 2^3\}$, $\{10, 9, 2^3, 1\}$, $\{10, 9, 2^3\}$, $\{10, 9, 2^3, 1\}$.

Also note that if B denotes one of these subsets of $\{1, \ldots, 10\}$, then $\text{prod}(B)^2$ divides $10!$ and $\text{prod}(B)$ is the maximum.

This proves that $\{5, 3, 6, 2^3\}$, $\{5, 3, 6, 2^3, 1\}$, $\{5, 3, 6, 2, 2^2\}$, $\{5, 3, 6, 2, 2^2, 1\}$, $\{5, 9, 2, 2^3\}$, $\{5, 9, 2, 2^3, 1\}$, $\{10, 3, 6, 2^2\}$, $\{10, 3, 6, 2^2, 1\}$, $\{10, 9, 2^3\}$, $\{10, 9, 2^3, 1\}$, $\{10, 9, 2^3\}$, $\{10, 9, 2^3, 1\}$ are precisely all the subsets of $\{1, \ldots, 10\}$ having the required property. Thus we could take $A = \{1, 2, 3, 4, 5, 6, 7\}, B = \{8, 9, 10\}$ for instance. ♣

Remark. Note that the above discussion provides more than what has been required. After observing that $\text{prod}(B)$ divides $2^4 \cdot 3^2 \cdot 5$, one may show that there is a subset B with $\text{prod}(B)$ equal to $2^4 \cdot 3^2 \cdot 5$ (for instance, $B = \{8, 9, 10\}$), and then conclude.

Solution 10. Let A, B be two nonempty disjoint subsets of X satisfying the required conditions (note that such subsets exist since X can be written as the

union of two disjoint subsets in finitely many ways only). Due to the equality

$$\frac{\text{prod}(A)}{\text{prod}(B)} = \frac{10!}{(\text{prod}(B))^2},$$

it is equivalent to having a subset B of X such that $\text{prod}(B)^2$ divides $10!$ and $\text{prod}(B)$ is the maximum. Note that $10!$ is equal to the product $2^8 \cdot 3^4 \cdot 5^2 \cdot 7$. So $\text{prod}(B)$ divides $2^4 \cdot 3^2 \cdot 5$. If $B = \{8, 9, 10\}$, then $\text{prod}(B)$ is equal to $2^4 \cdot 3^2 \cdot 5$. Hence, $A = \{1, \ldots, 7\}, B = \{8, 9, 10\}$ are two disjoint nonempty subsets of $X = \{1, \ldots, 10\}$ satisfying the required conditions. ■

Remark. Don't be surprised that it took a bit long to arrive at the above solution. It is often the case. Further, it is a standard practice to write down a complete solution as the final one, without any reference to the prior attempts (possibly several). Those attempts have their important role in providing insights, which may lead to a solution. Here, the details of those attempts have not been hidden from you, in order to take you along the journey. However, I would like to highlight that a *solution* to a problem has to be complete, and at the same time, has to be free from the prior thoughts that have no direct role to play in that solution, though they might have played a significant role in gaining insight.

Exercise 14 (RMO 2013c P6,). Let $n \geq 4$ be a natural number. Let $A_1A_2 \cdots A_n$ be a regular polygon and $X = \{1, 2, 3, \ldots, n\}$. A subset $\{i_1, i_2, \ldots, i_k\}$ of X, with $k \geq 3$ and $i_1 < i_2 < \cdots < i_k$, is called a *good subset* if the angles of the polygon $A_{i_1}A_{i_2} \cdots A_{i_k}$, when arranged in the increasing order, are in an arithmetic progression. If n is a prime, show that a proper good subset of X contains exactly four elements.

Walkthrough —

(a) Note that the angle subtended by any side of a regular n-gon at the center of its circumcircle is equal to $2\pi/n$.

(b) Show that the angle, subtended by any arc corresponding to two consecutive vertices of a polygon of the form $A_{i_1}A_{i_2} \ldots A_{i_k}$, is a multiple of $2\pi/n$.

(c) Conclude that the angles of a polygon of the form $A_{i_1}A_{i_2} \ldots A_{i_k}$ is a multiple of π/n.

Solution 11. Note that the angle subtended by any side of the regular n-gon $A_1A_2 \cdots A_n$ at the center of its circumcircle is equal to $2\pi/n$. Let $k \geq 3$ be a positive integer, and $\{i_1, i_2, \ldots, i_k\}$ be a proper good subset of $\{1, 2, \ldots, n\}$, with $i_1 < i_2 < \cdots < i_k$.

This shows that the angle, subtended by any arc corresponding to two consecutive vertices of the polygon $A_{i_1}A_{i_2} \ldots A_{i_k}$, is a multiple of $2\pi/n$. Since

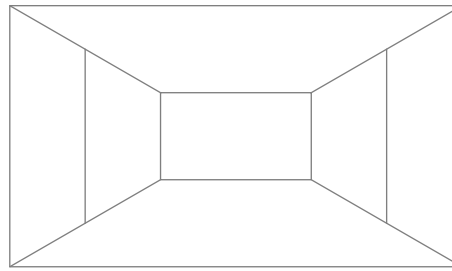


Figure 7: India RMO 2014d P6, Exercise 15

the angle subtended by an arc of a circle at its center, is twice of the angle subtended by it at any point on the circumference, it follows that the angles of the polygon are multiples of π/n. Since $\{i_1, i_2, \ldots, i_k\}$ is a good subset, and the angles of a k-gon add up to $(k-2)\pi$, it follows that there are k integers $n_1 \leq n_2 \leq \cdots \leq n_k$, which are in arithmetic progression, and their sum is equal to $(k-2)n$. This shows that $k(n_1+n_k)/2 = (k-2)n$, which implies that k divides $2(k-2)n$. Since $\{i_1, i_2, \ldots, i_k\}$ is a proper subset of $\{1, 2, \ldots, n\}$, and n is a prime, it follows that the integers k, n are relatively prime. This gives that k divides $2(k-2)$, and consequently, k divides 4. Using the bound $k \geq 3$, we obtain $k = 4$. ■

Exercise 15 (RMO 2014d P6,). In Fig. 7, can the numbers $1, 2, 3, 4, \ldots, 18$ be placed, one on each line segment, such that the sum of the numbers on the three line segments meeting at each point is divisible by 3?

> **Summary** — Since there are 18 line segments, it follows that if the integers $0, 1, 2$ can be put on the segments, using each of them exactly six times, such that 3 divides the sum of the integers on the segments meeting at any given point, then it would be possible to place $1, 2, \ldots, 18$ satisfying the required condition.

Solution 12. Note that if the integers $0, 1, 2$ can be put on the segments, using each of them exactly six times, such that 3 divides the sum of the integers on the segments meeting at any given point, then it would be possible to place $1, 2, \ldots, 18$ satisfying the required condition (by replacing the 0's (resp. 1's, 2's) by the six integers among $1, 2, \ldots, 18$ which are congruent to 0 (resp. $1, 2$) modulo 3, and such a replacement can be carried out since there are six elements among $1, 2, \ldots, 18$ congruent to $i \bmod 3$ for any $i \in \{0, 1, 2\}$). We now show that such an arrangement of $0, 1, 2$ exists. First, put 0's on all the vertical segments as in Fig. 8b, and then put 1's on the 'diagonal' segments as shown in Fig. 8c. This forces to put 2's on the 'diagonal' segments as shown in Figure 8d, which in turn, forces to write 1's and 2's on the horizontal segments as in Figure 8e. Note that the sum of the numbers (as in Figure 8e) on the

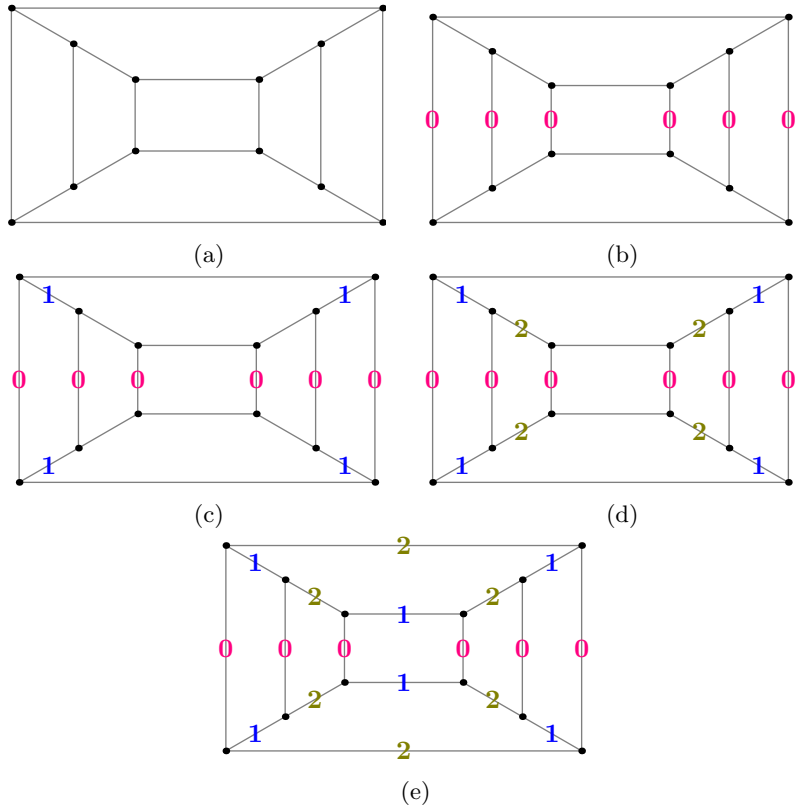


Figure 8: India RMO 2014, Exercise 15

three line segments meeting at each point is divisible by 3. So this gives an arrangement of $0, 1, 2$ satisfying the desired properties, then $1, 2, \ldots, 18$ can be arranged satisfying the given conditions (as described above). ■

Exercise 15 leads to the following question.

Question 16. Under which conditions, does a k-regular graph admit an edge coloring by the k-th roots of unity such that the sum of the colors incident at any vertex equals to zero?

Exercise 17 (RMO 2017b P1)**.** Consider a chessboard of size 8 units $\times$ 8 units (i.e. each small square on the board has a side length of 1 unit). Let S be the set of all the 81 vertices of all the squares on the board. What is the number of line segments whose vertices are in S, and whose length is a positive integer? (The segments need not be parallel to the sides of the board.)

Summary — A segment having vertices in S and length a positive integer, is horizontal or vertical, or the hypotenuse of a right-angled triangle whose smaller sides are parallel to the sides of the board. To count such right-angled triangles, note that they cannot have a too large hypotenuse.

Walkthrough —

(a) Determine the number of the horizontal segments with vertices in S and whose lengths are positive integers.

(b) By symmetry, the number of such vertical segments is equal to the above.

(c) To determine the slanted ones, note that such a slanted segment is the hypotenuse of a right-angled triangle whose smaller sides are parallel to the sides of the board, and have integer lengths. Note that the diagonal of an 8×8 chessboard has length $8\sqrt{2} < 12$. Thus, the only right-angled triangles, that can be fit within the board having sides parallel to the sides of the board and of integer length, have side lengths equal to $(3, 4, 5)$, $(4, 3, 5)$, $(6, 8, 10)$, $(8, 6, 10)$.

(d) Does a symmetry argument help? For instance, flipping around a diagonal, and then flipping around an axis (i.e. a line parallel to one of the sides of the board and dividing the board in two equal halves).

Solution 13. Note that within each horizontal line, there are $8-\ell+1$ horizontal segments of length ℓ for any $1 \leq \ell \leq 8$. This shows that the number of the horizontal segments with vertices in S and whose lengths are positive integers is equal to

$$8 \times (1 + 2 + \cdots + 8) = \frac{1}{2} 8^2 \cdot 9.$$

By symmetry, the number of such vertical segments is also equal to $\frac{1}{2} 8^2 \cdot 9$. Hence, there are $8^2 \cdot 9$ segments parallel to the sides of the board, which have vertices in S and whose lengths are positive integers.

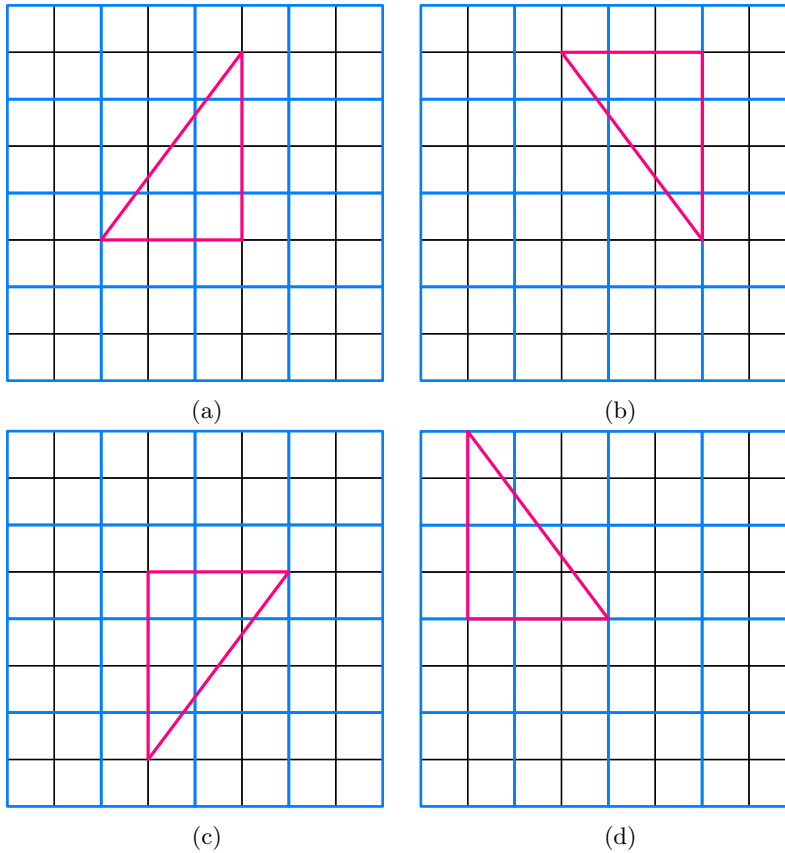


Figure 9: India RMO 2017b P1 — Several configurations of triangles with side lengths $(3, 4, 5)$, Exercise 17

Note that the diagonal of an 8×8 chessboard has length $8\sqrt{2} < 12$. Thus, the only right-angled triangles, that can be fit within the board having sides parallel to the sides of the board and of integer length, have the side lengths equal to $(3, 4, 5)$, $(4, 3, 5)$, $(6, 8, 10)$, $(8, 6, 10)$. The number of such right-angled triangles, having the side lengths equal to $(3, 4, 5)$, is equal to

$$4 \times (8 - 3 + 1) \times (8 - 4 + 1).$$

The number of such right-angled triangles, having the side lengths equal to the remaining triples, can be expressed in a similar way. This shows that the total count of such triangles is

$$2 \times 4 \times ((8 - 3 + 1) \times (8 - 4 + 1) + (8 - 6 + 1) \times (8 - 8 + 1)) = 360.$$

Hence, the number of the line segments with the stated property is

$$8^2 \cdot 9 + \frac{1}{2} \cdot 360 = 756.$$

■

Remark. Note that the above argument considers the line segments whose endpoints are distinct. There are 81 line segments having equal end-points and end-points lying in S.

Exercise 18 (RMO 2017b P3)**.** Let $P(x)$ and $Q(x)$ be polynomials of degree 6 and degree 3 respectively, such that

$$P(x) > Q(x)^2 + Q(x) + x^2 - 6 \text{ for all } x \in \mathbb{R}.$$

If all the roots of $P(x)$ are real numbers, then prove that there exist two roots of $P(x)$, say α, β such that $|\alpha - \beta| < 1$.

Solution 14. For any real number x, note that

$$P(x) > \left(Q(x) + \frac{1}{2}\right)^2 + x^2 - \frac{25}{4} \geq x^2 - \frac{25}{4}$$

holds. This implies that any root α of $P(x)$ satisfies $\alpha^2 < \frac{25}{4}$. Since $P(x)$ is a polynomial of degree 6 and all its roots are real, it follows that its six roots lie in the interval $(-5/2, 5/2)$, which is the union of the five intervals

$$(-5/2, -3/2), [-3/2, -1/2), [-1/2, 1/2), [1/2, 3/2), [3/2, 5/2).$$

Hence, there are two roots α, β of $P(x)$ such that $|\alpha - \beta| < 1$. ■

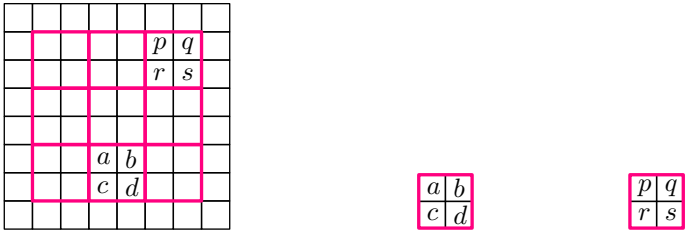


Figure 10: $ad - bc = \pm 1, ps - qr = \pm 1$, Exercise 19

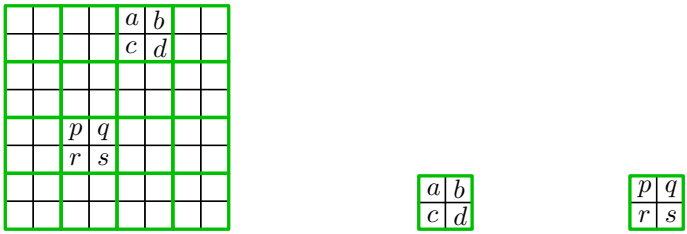


Figure 11: $ad - bc = \pm 1, ps - qr = \pm 1$, Exercise 19

Exercise 19 (Moscow Mathematical Olympiad 2015 Grade 11 Day 1 P5, proposed by O. N. Kosukhin)**.** Prove that it is impossible to put the integers from 1 to 64 (using each integer once) into an 8×8 table so that for any 2×2 square $\begin{array}{|c|c|}\hline a & b \\ \hline c & d \\ \hline \end{array}$, the difference $ad - bc$ is equal to 1 or -1.

Remark.

- Given a 2×2 square $\begin{array}{|c|c|}\hline a & b \\ \hline c & d \\ \hline \end{array}$, the difference $ad - bc$ is equal to

  the product of the diagonal terms
  − the product of the anti-diagonal terms.

  Let us call this difference *the determinant* of the 2×2 square.
- For instance,
  - $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ has determinant equal to -2,
  - $\begin{pmatrix} 8 & 9 \\ 7 & 12 \end{pmatrix}$ has determinant equal to $96 - 63 = 33$,
  - $\begin{pmatrix} 13 & 14 \\ 5 & 7 \end{pmatrix}$ has determinant equal to $91 - 70 = 21$.
  - Did you notice that if the determinant is odd, then the diagonal entries are odd or the anti-diagonal entries are odd?

- We need to show that there is no filling of an 8×8 table using the integers from 1 to 64, using each integer once, such that any 2×2 square (such squares have been marked in Fig. 10, Fig. 11, note that there $9 + 16 = 25$ such 2×2 squares.) has a determinant equal to 1 or -1. Equivalently[a], no matter how one may fill an 8×8 table using using the integers from 1 to 64, using each integer once, some 2×2 square has to have a determinant other than $1, -1$.

[a] Is the equivalence clear? Try to think about it!

Summary — If such a filling exists, then divide the 8×8 table into 16 pairwise disjoint 2×2 squares (as in Fig. 12). Due to parity constraints, each square contains precisely two evens along its diagonal or anti-diagonal, and their product is at most one more than the product of the odd entries. Consequently, for any of these 16 squares, the product of its even entries is less than the product of the successors of its odd entries. Multiplying across the squares gives a contradiction.

Walkthrough —

(a) Assume that such a filling exists.

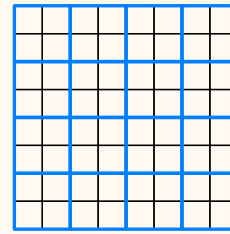


Figure 12: Moscow MO 2015 Grade 11 Day 1 P5, Exercise 19

(b) Recall that the determinant of a 2×2 square $\begin{array}{|c|c|}\hline a & b \\ \hline c & d \\ \hline \end{array}$ is

the product of the diagonal terms
$-$ the product of the anti-diagonal terms.

(c) Note that $\boxed{\text{even} - \text{even} \neq \pm 1, \text{odd} - \text{odd} \neq \pm 1}$, and hence any square contains two odd numbers along the diagonal  or on the anti-diagonal  .

(d) Divide the 8×8 table into 16 pairwise disjoint 2×2 squares.

(e) Each of these 16 squares contains at least two odd integers, and hence, they together contain at least 32 odd integers.

(f) Conclude that each of these 16 squares contains precisely two odd integers, and precisely two even integers.

(g) Consider a square among them. It is of the form

$$\begin{array}{|c|c|}\hline a & b \\ \hline c & d \\ \hline \end{array}$$ with a, d both odd, and b, c both even,

or of the form

$$\begin{array}{|c|c|}\hline b & a \\ \hline d & c \\ \hline \end{array}$$ with a, d both odd, and b, c both even.

(h) The product of its even entries is at most one more than the product of its odd entries.

(i) Note that for any two odd positive integers b, c, the inequality $bc + 1 < (b + 1)(c + 1)$ holds.

(j) This shows that

the product of two evens between 1 and 64
$<$ the product of
two (possibly different) evens between 1 and 64.

(k) Multiply all the even entries of the 16 squares to obtain

$$2 \cdot 4 \cdot \ldots \cdot 64 < (1 + 1) \cdot (3 + 1) \cdot \ldots \cdot (63 + 1) = 2 \cdot 4 \cdot \ldots \cdot 64.$$

Solution 15. Let us assume that an 8×8 table admits a filling by the integer from 1 to 64, using each integer once, such that each 2×2 square, considered as a matrix, has determinant equal to 1 or -1.

Claim — Any 2×2 square contains at least two odd integers.

Proof of the claim. Since the difference of two integers can be odd only when they are of different parity (i.e. one of them is odd, and the other is even), it follows that for any 2×2 square, the product of its diagonal entries and the product of its anti-diagonal entries are of different parity, and hence of these two products is odd, and consequently, the diagonal entries are odd or the anti-diagonal entries are odd. In particular, any 2×2 square contains at least two odd integers. □

Let us divide the 8×8 table into 16 pairwise disjoint 2×2 squares (as in Fig. 12).

Claim — Each of these 16 squares contains exactly two even integers, lying along its diagonal or anti-diagonal.

Proof of the claim. By the previous Claim, each of these 16 squares contains at least two odd integers, and they contain at least $16 \times 2 = 32$ odd integers. Since there are precisely 32 odd integers between 1 and 64, it follows that each of these 16 squares contains exactly two odd integers along its diagonal or anti-diagonal, and hence exactly two even integers along its anti-diagonal or diagonal. □

Since the determinant of any 2×2 square is 1 or -1, it follows that for any of the 16 squares as in Fig. 12, the product of its even entries is at most one more than the product of its odd entries. Note that for any two odd positive integers b, c, the inequality $bc + 1 < (b+1)(c+1)$ holds. Consequently, for any of the 16 squares as in Fig. 12, the product of its even entries is less than the product of the successors of its odd entries. This implies that the product of the even entries of all the 16 squares is less than the product of the successors of the odd entries of these boxes. Note that the even entries of these squares are the even integers lying between 1 and 64, so are the successors of the odd entries of these squares. It follows that

$$2 \cdot 4 \cdot \ldots \cdot 64 < (1+1) \cdot (3+1) \cdot \ldots \cdot (63+1) = 2 \cdot 4 \cdot \ldots \cdot 64.$$

This contradicts the assumption that an 8×8 table admits a filling by the integer from 1 to 64, using each integer once, such that each 2×2 square, considered as a matrix, has determinant equal to 1 or -1. Hence, no such filling is possible. ■

Exercise 20 (USAMO 2000 P4,). **Find the smallest positive integer n such that if n squares of a 1000×1000 chessboard are colored, then there will exist three colored squares whose centers form a right triangle with sides parallel to the edges of the board.**

Walkthrough —

(a) Does Fig. 13 help?

(b) Consider a suitable coloring to show that $n \geq 1999$.

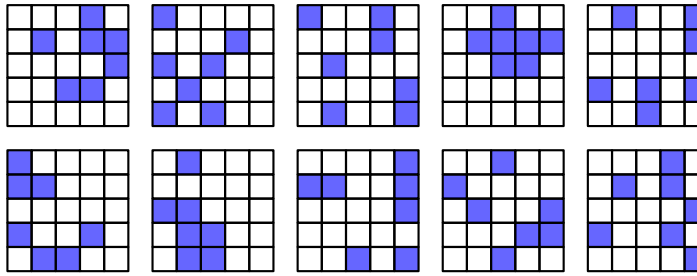
(c) Does coloring 1999 squares suffice?

(d) What happens when there are two columns, each containing at least two colored squares, and some row contains one colored square from each of these columns?

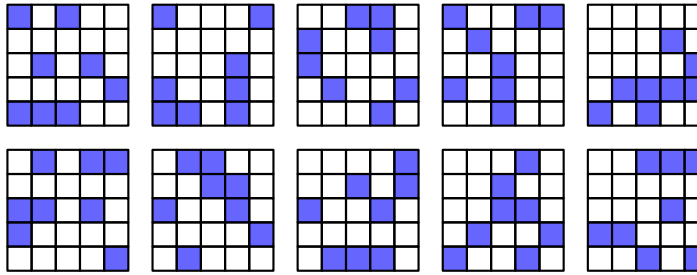
(e) Consider the columns which contain at least two colored squares. How many colored squares can they contain together?

(f) What about the remaining colored squares? How many are they in total?

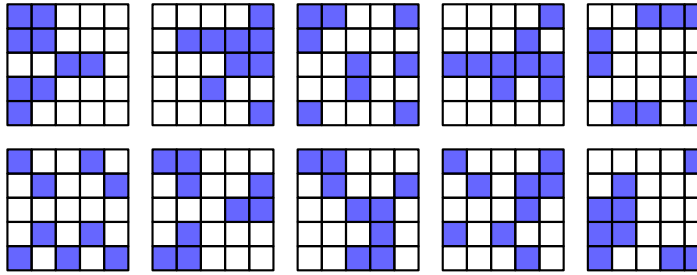
Solution 16. Note that if all the squares on the top row and on the left column, excluding the square common to them, are colored, then the required condition is not satisfied. This shows that n is at least 1999.



The above chessboards have 7 of its squares colored



The above chessboards have 8 of its squares colored



The above chessboards have 9 of its squares colored

Figure 13: USAMO 2000 P4, Exercise 20

Suppose precisely 1999 squares of the chessboard have been colored. Note that there are columns containing at least two colored squares. Consider the columns containing at least two colored squares. If there are two among these columns, such that some row contains one colored square from each of these columns, then we are done. It remains to consider the case that no two of these columns have this property, which we assume from now on. Hence, the colored squares lying on these columns, lie in different rows. This shows that these columns together contain at most 1000 colored squares. Consequently, there are at least 999 colored squares, each lying on a colum containing no more colored squares. This implies that there are precisely 999 such colored squares. Moreover, there is only one column containing at least two colored squares, and it contains the remaining 1000 colored squares. One among the 999 colored squares, together with two suitable squares from these 1000 squares, have the required property. This shows that $n = 1999$. ■

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